Physics - Undergraduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T. The average total energy of the oscillator is
   (a) (1/2) k T
   (b) kT
   (c) (3/2) k T
   (d) 3 kT

2. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
   (a) 1.2 x 10^9 m/s
   (b) 3.0 x 10^8 m/s
   (c) 1.5 x 10^8 m/s
   (d) 1.0 x 10^8 m/s

3. The quantum efficiency of a photon detector is 0.1. If$_{100}$ photons are sent into the detector, one after the other, the detector will detect photons
   (a) exactly 10 times
   (b) an average of 10 times, with an rms deviation of about 0.1
   (c) an average of 10 times, with an rms deviation of about 1
   (d) an average of 10 times, with an rms deviation of about 3

4. Which of the following ions CANNOT be used as a dopant in germanium to make an n-type semiconductor?
   (a) As
   (b) P
   (c) Sb
   (d) B

5. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
   (a) Constant temperature
   (b) Constant volume
   (c) Constant pressure
   (d) Adiabatic

True / False (10 marks)

Answer True or False

6. True or False: The electric displacement current through a surface is proportional to the rate of change of the magnetic flux through that surface.

7. True or False: The electric displacement current through a surface S is proportional to the rate of change of the magnetic flux through S.

8. True or False: The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately 1.0 GeV/c^2.

9. True or False: Doubling the voltage across a resistor results in a new rate of energy dissipation of 2 W.

10. True or False: If a particle's total energy is twice its rest energy, its relativistic momentum is equal to mc/2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the impact of closing one end of an organ pipe on the harmonics produced. Which harmonics from an open pipe will be affected, and why?

12. Discuss how the speed of sound changes with temperature and calculate the resonant frequency of a pipe open at both ends on a cold day when the speed of sound is reduced by 3%.

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